Product Analyst Home Assignment

Part 2 - Product Review

Part A: Product Review

The PRD defines the main reporting flows and release scope, but several areas require clarification before implementation. I prioritized the following five gaps because they have the highest potential impact on security, report reliability, and the customer experience, and could affect the feature's behavior or scope if left unresolved.

1. Report Data Permissions / RBAC - Critical

What is missing or unclear:
The PRD states that report objects are governed by the platform's RBAC permissions for create, edit, and delete actions, but it does not define which user's permissions determine the data included when a report is generated. This is especially unclear for organization-level and scheduled reports, where the report creator, workflow owner, user triggering the report, and recipient may be different users.

Why it matters:
Reports may contain sensitive security data. Without a defined permission context, a generated report could potentially include data that the person accessing it would not normally be allowed to see.

What I would need to know:
Which user's RBAC scope should apply when a report is generated? How should permission changes after a report or Workflow is created affect future report generation and access?

2. Report and Dependency Lifecycle - High

What is missing or unclear:
Reports can depend on Dashboards or Saved Views and can also be connected to active Workflows. The PRD explains how dashboard changes can sync to future reports, but it does not define what happens when a dependency is deleted or becomes unavailable. For example, it is unclear what happens if a source Dashboard or Saved View is deleted, or if a Report used by an active Workflow is deleted.

Why it matters:
Without defined dependency behavior, reports may become invalid or active Workflows may remain connected to reports that can no longer be generated correctly, resulting in missing or unexpected scheduled reports.

What I would need to know:
What should happen when a source Dashboard or Saved View is deleted or becomes unavailable? What should happen to an active Workflow when its linked Report is edited or deleted?

3. Scheduled Report Recipient and Access Rules - High

What is missing or unclear:
The PRD supports scheduled delivery by email and states that when a report is too large to be attached, the recipient receives an authenticated download link. However, it does not define who can be a report recipient or what access requirements apply to that recipient.

Why it matters:
It is unclear whether reports can be sent to any email address or only to users within the Upwind organization. This also affects the large-file flow: a recipient who does not meet the required authentication requirements may receive the link but be unable to access the report.

What I would need to know:
Who can be selected as a scheduled report recipient? Do recipients need an Upwind account or specific permissions? How should authentication work for recipients receiving a download link?

4. Scheduled Report Data Freshness - High

What is missing or unclear:
The PRD supports scheduled delivery for saved Reports through Workflows, but it does not explicitly define whether each scheduled execution regenerates the Report using the latest underlying data. For Summary Reports, the PRD defines how source Dashboard configuration changes can sync to future Reports, but this does not clarify the freshness of the underlying security data. For Operational Data Reports, no equivalent data-freshness behavior is defined.

Why it matters:
Scheduled Reports are intended to support recurring security reporting. If data freshness is not clearly defined, recipients may receive Reports that do not reflect the expected point in time, reducing the reliability of automated reporting.

What I would need to know:
Does every scheduled execution regenerate the Report using the latest available underlying data? Are there any snapshot, caching, or data-freshness rules that apply to scheduled Report generation?

5. Delivery Channel Scope Is Inconsistent - Medium

What is missing or unclear:
The Create Report mockups state that automated delivery is available through email, Slack, or a ticketing system. However, the release capabilities specify

automated delivery by email, and the Workflow mockup also shows an email delivery action. It is therefore unclear whether Slack and ticketing integrations are part of the initial release.

Why it matters:
The inconsistency creates different expectations between the customer-facing UI and the defined release scope. If the initial release is email-only, the current UI copy describes capabilities that will not be available.

What I would need to know:
Which delivery channels are included in the initial release? If V1 is email-only, the Create Report copy should be updated. If Slack and ticketing integrations are included, they should be explicitly reflected in the release requirements and Workflow configuration.

Part B: Test Plan

Test cases / acceptance criteria

1. Summary Report (PDF) – Creation & Generation

Test Case: Create a Summary Report from a Predefined Dashboard
Expected Result: The report can be configured successfully using the selected Predefined Dashboard.
Priority: Critical

Test Case: Create a Summary Report from a Custom Dashboard
Expected Result: The report can be configured successfully using the selected Custom Dashboard.
Priority: Critical

Test Case: Generate and download a Summary Report on demand
Expected Result: A valid PDF is generated and downloaded successfully.
Priority: Critical

Test Case: Verify generated PDF content
Expected Result: The PDF reflects the selected Dashboard, configured scope, and relative time range.
Priority: Critical

Test Case: Save a Summary Report for future use
Expected Result: The report is saved successfully and appears on the Reports page with its configured details.
Priority: High

Test Case: Generate a previously saved Summary Report
Expected Result: A PDF can be generated successfully using the saved report configuration.
Priority: High

2. Operational Data (CSV) – Creation & Generation

Test Case: Create an Operational Data report from a supported security module
Expected Result: The report can be configured successfully using the selected module/data source.
Priority: Critical

Test Case: Select and order CSV columns
Expected Result: The generated CSV contains the selected columns in the configured order.
Priority: Critical

Test Case: Apply scope filters
Expected Result: The CSV contains only records matching the configured scope

and filters.
Priority: Critical

Test Case: Generate and download an Operational Data report on demand
Expected Result: A valid CSV is generated and downloaded successfully.
Priority: Critical

Test Case: Verify exported data correctness
Expected Result: The CSV contains the expected raw findings according to the selected module, fields, and scope.
Priority: Critical

Test Case: Save an Operational Data report
Expected Result: The report is saved successfully and appears on the Reports page.
Priority: High

Test Case: Generate a previously saved Operational Data report
Expected Result: A CSV can be generated using the saved report configuration.
Priority: High

3. Scope, Filters & Time Range

Test Case: Generate a report with a defined scope
Expected Result: The generated report contains only data matching the selected scope.
Priority: Critical

Test Case: Use an existing Saved View as the report scope
Expected Result: The Saved View's filtering criteria are applied to the report.
Priority: High

Test Case: Apply a relative time range to a Summary Report
Expected Result: The generated PDF reflects the configured relative time period.
Priority: Critical

4. Report Management & Visibility

Test Case: View a saved Report on the Reports page
Expected Result: The saved Report appears with the correct details.
Priority: High

Test Case: Search and filter Reports
Expected Result: Only Reports matching the selected search/filter criteria are displayed.
Priority: Medium

Test Case: Download a Report from the Reports page
Expected Result: The selected Report can be downloaded successfully.
Priority: High

Test Case: Edit a saved Report
Expected Result: The permitted changes are saved and reflected on the Report object.
Priority: High

Test Case: Delete a saved Report
Expected Result: The Report is removed and is no longer available from the Reports page.
Priority: High

Test Case: Create a Report with "Only for me" visibility
Expected Result: The Report is not visible to other organization users.
Priority: Critical

Test Case: Create a Report with "Organization level" visibility
Expected Result: The Report is saved with "Organization level" visibility and is available for organization-level sharing.
Priority: Critical

Test Case: Verify linked Workflow information
Expected Result: A Report associated with a Workflow displays the relevant Workflow association.
Priority: Medium

5. Save as Custom Report

Test Case: Open "Save as Custom Report" from a supported Board view
Expected Result: The report creation flow opens from the selected Board.
Priority: High

Test Case: Open "Save as Custom Report" from a supported table/list view
Expected Result: The report creation flow opens from the selected table/list view.
Priority: High

Test Case: Verify configuration is pre-populated
Expected Result: Relevant configuration from the source view is carried into the creation flow.
Priority: High

Test Case: Complete and save the pre-populated Report
Expected Result: The Report is created successfully with the pre-populated settings and the entered name, description, and visibility level.
Priority: High

6. Scheduled Delivery & Workflows

Test Case: Connect a saved Report to a Workflow
Expected Result: The selected Report is successfully associated with the Workflow.
Priority: Critical

Test Case: Run the Workflow according to its configured schedule
Expected Result: The scheduled Report delivery is triggered at the configured schedule.
Priority: Critical

Test Case: Deliver a scheduled Report by email
Expected Result: The Report is delivered to the configured email recipient(s).
Priority: Critical

Test Case: Deliver a Report within the destination size limit
Expected Result: The Report is delivered as an email attachment.
Priority: High

Test Case: Deliver a Report exceeding the destination size limit
Expected Result: A link to the Report is sent instead of an attachment, and accessing the Report through the link requires authentication.
Priority: High

7. RBAC / Report Object Permissions

Test Case: Create a Report with the required RBAC permission
Expected Result: The authorized user can create the Report.
Priority: High

Test Case: Attempt to create a Report without the required RBAC permission
Expected Result: The user is prevented from creating the Report.
Priority: Critical

Test Case: View a Report with the required RBAC permission
Expected Result: The authorized user can access the Report object.
Priority: High

Test Case: Attempt to view a Report without the required RBAC permission
Expected Result: The user is prevented from accessing the Report object.
Priority: Critical

Test Case: Edit a Report with the required RBAC permission
Expected Result: The authorized user can edit the Report.
Priority: High

Test Case: Attempt to edit a Report without the required RBAC permission
Expected Result: The user is prevented from editing the Report.
Priority: Critical

Test Case: Delete a Report with the required RBAC permission
Expected Result: The authorized user can delete the Report.
Priority: High

Test Case: Attempt to delete a Report without the required RBAC permission
Expected Result: The user is prevented from deleting the Report.
Priority: Critical

8. Audit Logging

Test Case: Create a Report
Expected Result: The create action is recorded in the audit log.
Priority: High

Test Case: Edit a Report
Expected Result: The update action is recorded in the audit log.
Priority: High

Test Case: Delete a Report
Expected Result: The delete action is recorded in the audit log.
Priority: High

9. Sync Board Changes

Test Case: Modify a Custom Dashboard after enabling "Sync board changes"
Expected Result: Future generated Reports reflect the updated Board configuration.
Priority: Critical

Edge cases

The following edge cases have a defined expected behavior in the PRD or supporting mockups and can therefore be validated in this testing round.

1. Create a Report without a description
Test Case: Create and save a Report without entering a description.
Expected Result: The Report is created successfully without a description.
Priority: Medium

2. Generate an Operational Data Report at the CSV record limit
Test Case: Generate an Operational Data Report containing exactly 500K records.
Expected Result: The CSV is generated successfully with 500K records.
Priority: High

3. Sync multiple Dashboard changes before the next generation
Test Case: Make multiple changes to a Custom Dashboard after enabling "Sync board changes", then generate the Report.
Expected Result: The newly generated Report reflects the latest source

Dashboard configuration, including the changes made before generation.
Priority: High

What you would explicitly exclude from this round of testing?

Explicitly Excluded from This Testing Round

The following scenarios are excluded from this testing round because their expected behavior is not fully defined in the current PRD, and additional specification is required before they can be tested.

1. **Source dependency lifecycle** - The PRD does not define what happens when a source Dashboard or Saved View used by a Report is later deleted or becomes unavailable, including the impact on future Report generations and associated Workflows.
2. **Dashboard changes when Sync is disabled** - It is unclear whether a Report should preserve its original Dashboard configuration or reflect later changes when “Sync board changes to future reports” is disabled.
3. **Reports with no matching data** - The expected behavior for Summary PDF or Operational Data CSV Reports when the selected scope or filters return no matching data is not defined.
4. **Operational Data exceeding 500K records** - The supported limit is defined, but the expected behavior when the result exceeds 500K records is not specified.
5. **Editing pre-populated settings in “Save as Custom Report”** - The PRD does not clarify whether pre-populated settings can be modified or removed before saving the Report.
6. **Report deletion while linked to an active Workflow** - The expected behavior of the associated Workflow after its Report is deleted is not defined.
7. **Scheduled execution failure** - Retry, failure status, and notification behavior for Report generation or email delivery failures are not defined.
8. **Scheduled Report data freshness** - The PRD does not clearly define whether each scheduled execution regenerates the Report using the latest underlying data.
9. **Secure link access and external recipients** - The PRD requires authentication for large-file download links but does not define recipient eligibility, access permissions, link expiration/reuse, or authentication behavior for recipients without platform access.

10. **RBAC and permission lifecycle** - The PRD does not fully define how Report visibility interacts with RBAC, what happens when permissions change after Report creation, or whose permissions and data-level access are applied during scheduled background execution.

11. **Audit logging beyond CRUD** - Audit logging is defined for create, update, and delete, but it is not specified whether Report generation, download, or scheduled delivery should also be logged.

12. **Operational Data in-app table behavior** - In-app tables are listed as part of Operational Data, but their creation, display, interaction, and relationship to CSV Reports are not defined.

Non-Functional Testing - Out of Scope for This Round

Report generation performance and concurrency limits - Performance, capacity, timeout, and concurrency expectations for large or simultaneous Report generations are not specified.

Part C - BI & Monitoring Definition

Adoption & Usage Metrics

1. Organization Adoption Rate

What it measures:
The percentage of organizations eligible to use Custom Reporting that have created at least one Report.

Why it matters:
Shows how broadly the feature is being adopted across the customer base. Measuring adoption at the organization level avoids overstating adoption when a small number of organizations create many Reports.

How to calculate:
Divide the number of eligible organizations that created at least one Report by the total number of organizations eligible to use Custom Reporting during the measured period.

2. Active Reporting Organizations

What it measures:
The percentage of organizations that adopted Custom Reporting and actively use the feature during a defined period.

Why it matters:
Shows whether Custom Reporting continues to provide value after initial adoption rather than being tried once and abandoned.

How to calculate:
Divide the number of organizations with at least one meaningful reporting activity during the period by the number of organizations that have adopted Custom Reporting. Meaningful activity includes creating a Report, generating a Report on demand, or having a scheduled Report executed. Break down active usage by Report type and execution type (on-demand vs. scheduled) to understand how active organizations use the feature.

3. Scheduled Reporting Adoption

What it measures:
The percentage of active reporting organizations using scheduled Report execution and delivery through Workflows.

Why it matters:
Shows whether customers are adopting the automation capability intended to reduce repetitive manual reporting and support recurring reporting workflows.

How to calculate:
Divide the number of active reporting organizations with at least one Report connected to an active scheduled Workflow by the total number of active reporting organizations. Break down the metric by Report type (Summary vs. Operational Data) to identify which reporting use cases are more commonly automated.

4. "Save as Custom Report" Adoption

What it measures:
The percentage of active reporting organizations using the "Save as Custom Report" capability from existing export points.

Why it matters:
Shows whether customers use the in-context creation flow designed to turn existing exports into reusable Reports, rather than requiring them to configure a Report from scratch.

How to calculate:
Divide the number of active reporting organizations that used "Save as Custom Report" at least once during the measured period by the total number of active reporting organizations. Break down usage by source/export point to identify where this entry point is most commonly used.

5. Report Type Adoption

What it measures:
The percentage of active reporting organizations using each Report type - Summary Reports and Operational Data Reports.

Why it matters:
Shows which reporting use cases customers are adopting and helps Product understand whether both Report types are providing value.

How to calculate:
For each Report type, divide the number of active reporting organizations that generated at least one Report of that type during the measured period by the total number of active reporting organizations. An organization may use both Report types, so the percentages do not need to add up to 100%.

Quality & Reliability Metrics

1. Report Generation Success Rate

What it measures:
The percentage of Report generation attempts that complete successfully.

Why it matters:
Report generation is the core functionality of Custom Reporting. A decline in

success rate directly prevents customers from receiving the Reports they expect and may indicate a problem in the generation pipeline.

How to calculate:
Divide the number of successfully completed Report generations by the total number of Report generation attempts during the measured period. Break down the metric by Report type (Summary vs. Operational Data), execution type (on-demand vs. scheduled), and failure reason to identify where generation problems occur.

2. Scheduled Workflow Execution Success Rate

What it measures:
The percentage of scheduled Report Workflow executions that run successfully when they are expected to run.

Why it matters:
A scheduled Report may fail before Report generation even begins. Tracking scheduled execution separately helps detect silent failures where customers expect an automated Report, but the Workflow never executes as planned.

How to calculate:
Divide the number of scheduled Report Workflow executions that ran successfully by the total number of scheduled Report Workflow executions expected during the measured period. Track missed executions and break down failures by failure reason to identify scheduling or Workflow-specific issues.

3. Scheduled Delivery Success Rate

What it measures:
The percentage of scheduled Report delivery attempts that are successfully delivered to their intended recipients.

Why it matters:
A Report can be generated successfully but still fail during delivery. Since scheduled reporting is intended to provide Reports automatically, delivery reliability directly affects the value customers receive from the feature.

How to calculate:
Divide the number of successful scheduled Report deliveries by the total number of scheduled Report delivery attempts during the measured period. Break down the metric by delivery method (attachment vs. secure download link) and failure reason to identify where delivery problems occur.

4. Report Generation Latency

What it measures:
The time between the start and completion of Report generation, with P95 latency used to monitor the slower end of the user experience.

Why it matters:
Even when Reports are generated successfully, slow generation can create a poor experience for on-demand users and cause delays or processing bottlenecks for scheduled Reports.

How to calculate:
Measure the duration between the start and completion of each successful Report generation and track the P95 generation time during the measured period. Break down latency by Report type, execution type (on-demand vs. scheduled), and Report size or record volume to distinguish expected differences in processing time from performance degradation.

Alerting Strategy

Operational issues that may prevent customers from generating or receiving Reports should trigger automatic alerts, while product adoption and usage trends can generally be reviewed weekly.

Automatic Alerts

- **Report Generation Success Rate:** Notify the team if the success rate drops below **98% over a rolling 15-minute period**, assuming sufficient Report volume.
- **Scheduled Workflow Execution Success Rate:** Notify the team if the success rate drops below **98% over a rolling 30-minute period**, or if multiple expected scheduled executions are missed.
- **Scheduled Delivery Success Rate:** Notify the team if the success rate drops below **98% over a rolling 30-minute period**, monitored separately by delivery method (attachment vs. secure download link).
- **Report Generation Latency:** Notify the team if P95 generation latency exceeds **120 seconds for a sustained 15-minute period**.
- **Significant Active Usage Drop:** Notify the Product team if active reporting organizations fall more than **30% below their comparable recent baseline**. This should trigger investigation rather than immediate on-call paging.

Weekly Review

Adoption and usage metrics — **Organization Adoption Rate, Active Reporting Organizations, Scheduled Reporting Adoption, "Save as Custom Report" Adoption, and Report Type Adoption** — should primarily be reviewed weekly to identify trends and changes in customer behavior. Quality and reliability metrics should also be reviewed over time to detect gradual degradation that does not cross an alert threshold.

Threshold Calibration

These are initial monitoring thresholds and should be calibrated after launch based on observed baseline performance, traffic volume, and agreed service-level objectives.

Additional Data / Instrumentation Needed

The PRD defines audit logging for Report creation, updates, and deletion, but it does not specify all of the telemetry required to calculate the metrics and alerts above. I would need the following data, unless it is already available elsewhere in the platform:

1. Report Usage and Creation Context

- Report ID, organization ID, Report type (Summary / Operational Data), creation timestamp, and creation entry point (Reports Page / "Save as Custom Report").
- For "Save as Custom Report", the source/export point should also be captured.

2. Report Generation Telemetry

- Generation start and completion timestamps, generation status (success / failure), trigger type (on-demand / scheduled), and failure reason.
- Report size and record count where relevant.

3. Scheduled Workflow Execution Telemetry

- Workflow ID, linked Report ID, Workflow status (active / inactive), expected execution time, actual execution time, execution status, and failure reason.

4. Scheduled Delivery Telemetry

- Delivery attempt timestamp, delivery outcome (success / failure), delivery method (attachment / secure download link), and failure reason.